Section D Comprehensive Review - Mixed

Directions: Solve each exponential equation. Show all work. No calculators.

1. Solve $3^x = \frac{1}{27}$.

2. Solve $5^{2x-2} = 625$.

3. Solve $4^{3x+2} = 4^{x+10}$.

4. Solve $2 \cdot 4^x = 128$.

5. Solve $10^x = \frac{1}{1000}$.

6. Solve $7 \cdot 2^x = 224$.

7. Solve $2^{3x+1} = 128$.

8. Solve $3 \cdot 5^x = 375$.

9. Solve $2^{6x+1} = 2^{x+10}$.

10. Solve $5^{2x-7} = 125$.

11. Solve $5^{4x-1} = 5^{x+8}$.

12. Solve $2^x = 256$.

13. Solve $2 \cdot 3^x = 162$.

14. Solve $5^x = 625$.

15. Solve $2^x = \frac{1}{64}$.

16. Solve $4 \cdot 5^x = 500$.

17. Solve $4^{2x+2} = 256$.

18. Solve $5^x = \frac{1}{125}$.

19. Solve $3^{3x-1} = 243$.

20. Solve $5^{x+2} = 3125$.

21. Solve $3^{5x-2} = 3^{x+7}$.

22. Solve $4 \cdot 2^x = 256$.

23. Solve $4^x = 256$.

24. Solve $3^{3x-4} = 243$.

25. Solve $6 \cdot 2^x = 192$.

26. Solve $2^x = 512$.

27. Solve $8 \cdot 2^x = 512$.

28. Solve $2^{x+4} = 256$.